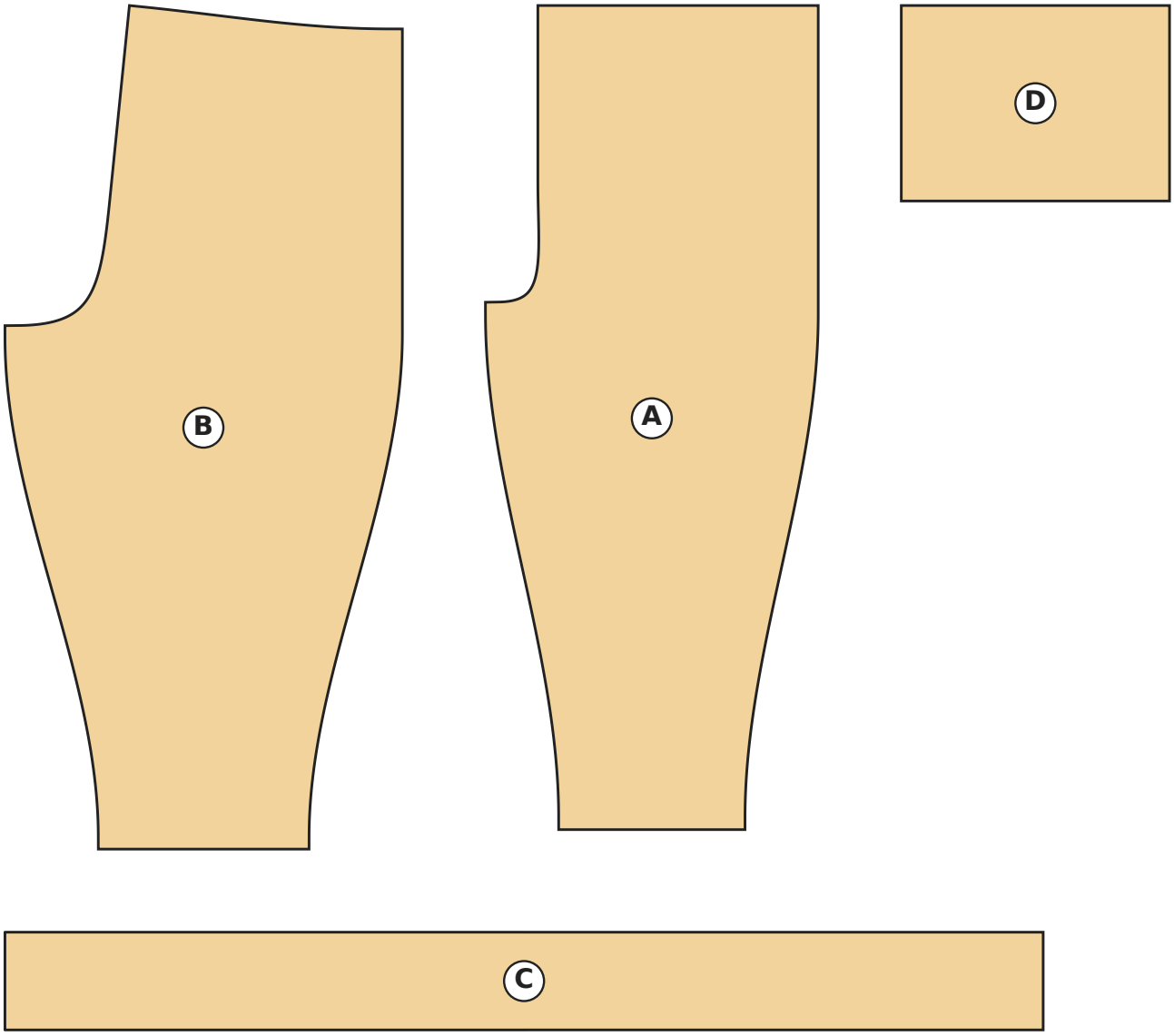
HAREM PANTS

size 122

harem (wide, dropped crotch) fit, Sweat / French Terry (25 % stretch)



4 parts · 20 steps

Read the steps once before you start. Every picture is drawn from your pattern pieces.

right side · wrong side · stitch · pin · notch: match

Parts



A 2×
Front leg



B 2×
Back leg



C 1×
Waistband (elastic)
cut on fold



D 2×
Leg cuff

Fabric and notions

- Main fabric: 0.90 m at 150 cm width
- Rib knit: 0.25 m at 70 cm width
- Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)
- Polyester sewing thread to match

Machines and tools



Sewing machine
narrow zigzag 1 × 2.5 mm (or overlocker); wide zigzag 4 × 2.5 mm; twin needle 4 mm, stitch 3 mm



Iron
steam



Pins or clips



Fabric scissors



Tape measure



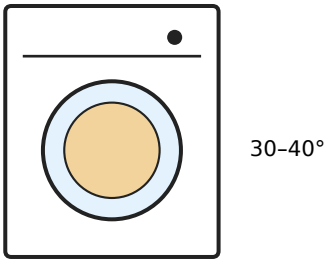
Chalk or a fabric pen



Safety pin

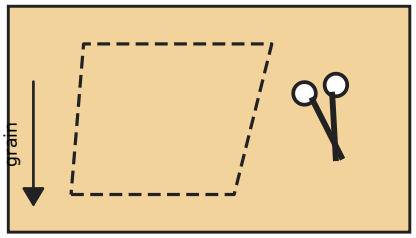
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

1 PREPARE
Pre-wash the fabric



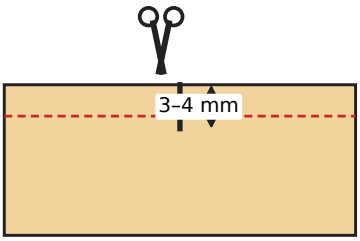
Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.

2
Cut out all pieces



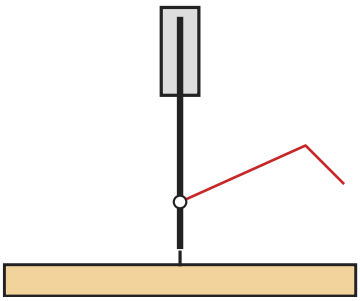
Follow the cutting layout; grainline arrows parallel to the selvage.

3
Snip the notches



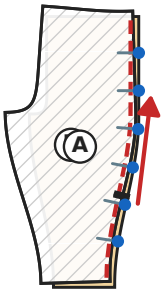
Cut 3–4 mm into the seam allowance at every notch.

4
Set up the machine



Jersey/stretch needle 75–80, polyester thread. Sew a test seam on a scrap first.

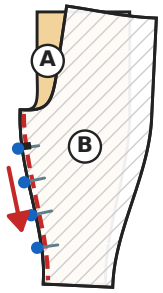
5 LEGS
Sew the side seams
×2



Front and back leg right sides together; single notches = front, double = back.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

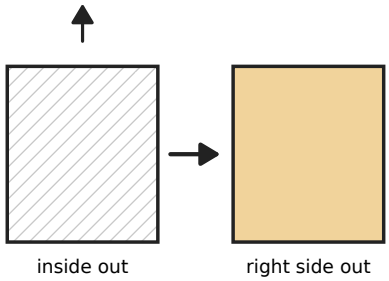
6
Sew the inseams
×2



The notches on the inseam meet.

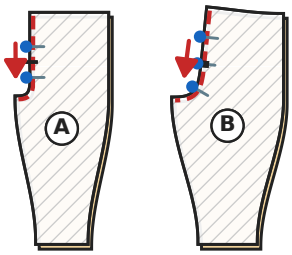
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

7 CROTCH
Put one leg inside the other



Turn one leg right side out and slip it into the other (wrong side out): the legs lie right sides together.

8
Sew the crotch seam

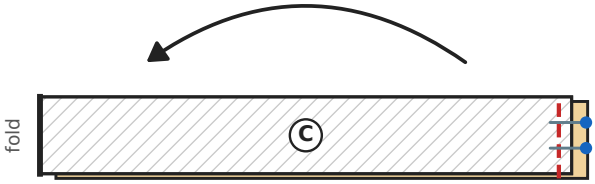


! Sew the crotch curve a second time for strength.

In one go from the front to the back top edge; the inseams meet exactly.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

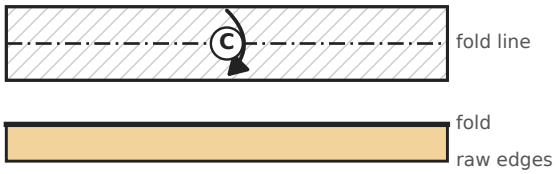
9 WAISTBAND
Close the waistband into a ring



Right sides together across the short ends (centre back); leave 3 cm open for the elastic.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

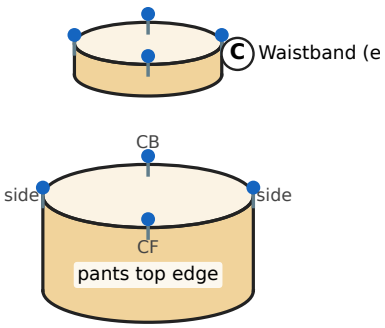
10
Fold the waistband in half



Lengthwise, wrong sides together, along the fold line.

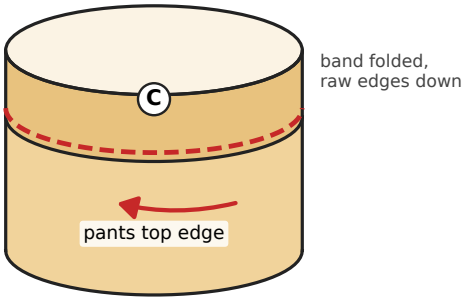
Iron: steam

11
Mark quarters



On waistband and pants top edge: centre front, centre back, sides.

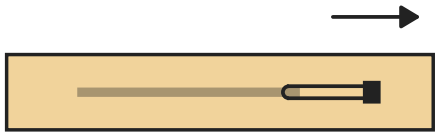
12
Sew the waistband on



Raw edges together, opening at the back on the inside; quarter marks meet.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

13 Thread the elastic

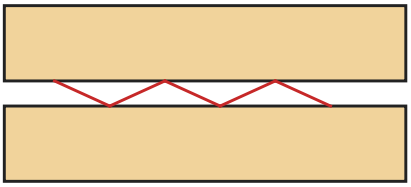


elastic, with a safety pin

With a safety pin through the opening; overlap the ends by 2 cm and sew them together.

Sewing machine: wide zigzag 4 × 2.5 mm

14 Close the opening



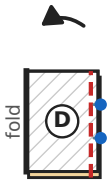
ladder stitch: the stitches disappear

Spread the gathers evenly, close the opening and stitch in the ditch at the side seams so the elastic cannot twist.

Sewing machine: twin needle 4 mm, stitch 3 mm

15 Close the cuffs into rings ×2

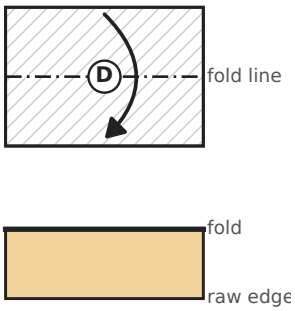
HEMS



Right sides together across the short ends.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

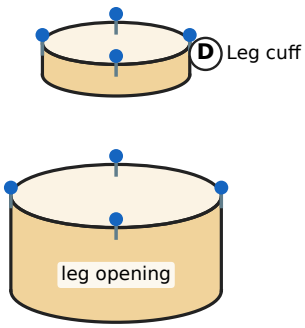
16 Fold the cuffs in half ×2



Lengthwise, wrong sides together.

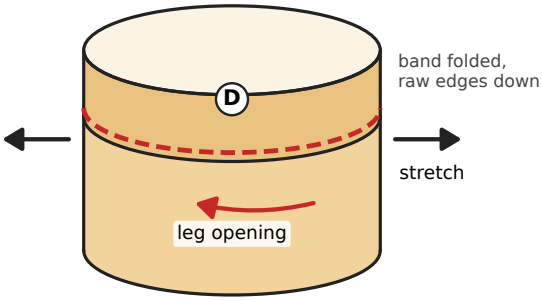
Iron: steam

17 Mark quarters ×2



On cuff and leg opening.

18 Sew the cuffs on ×2

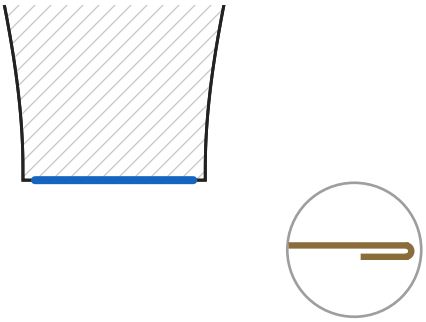


Slip the cuff over the right-side-out leg, cuff seam at the inseam.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

19 Press the allowance up

×2



Seam allowance towards the pants.

Iron: steam

20 FINISH Done!



Press all seams with steam; do not press down hard on sweat.

Harem pants · size 122

harem (wide, dropped crotch) fit, Sweat / French Terry (25 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

Files

- PDF roll 610 mm (600 × 1062 mm): `harem_pants_size122_roll610.pdf`
- Preview roll 610 mm: `harem_pants_size122_roll610_preview.png`
- SVG master 1:1: `harem_pants_size122_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

Body measurements

Body measurement	Value	Source
Body height	122.0 cm	table
Chest	62.0 cm	table
Waist	56.0 cm	table
Hip	66.0 cm	table
Head	52.0 cm	table
Neck	28.5 cm	table
Inseam	54.0 cm	table
Side length	72.0 cm	table
Arm length	47.0 cm	table
Back length	29.5 cm	table
Shoulder width	10.0 cm	table
Thigh	37.0 cm	table
Ankle	17.8 cm	table

Finished measurements (virtually sewn)

Finished measurement	Value
Hip width	85.8 cm
Thigh width	55.5 cm
Leg width at mid-leg	42.0 cm
Leg opening	20.0 cm
Waistband relaxed	50.4 cm
Waistband stretched (pants top edge)	83.0 cm
Outseam	75.5 cm
Inseam	48.2 cm
Front rise	27.3 cm
Back rise	29.1 cm
Waistband height	3.0 cm
Cuff height	7.0 cm

Finished measurement	Value
Crotch seam total	64.2 cm

Pattern pieces

Piece	Cutting	Fabric	Seam allowance
Front leg	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Back leg	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Waistband (elastic)	cut 1 on fold	Main fabric	seam allowance 1.0 cm included
Leg cuff	cut 2	Rib knit	seam allowance 1.0 cm included

Fabric needed

- **Main fabric:** about 0.90 m at 150 cm width (folded selvage to selvage, cut on fold pieces at the fold, incl. 10 % for shrinkage; single layer would need 0.90 m)
- **Rib knit:** about 0.25 m at 70 cm width (single layer, incl. 10 % for shrinkage)
- Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

Cut on fold (Waistband (elastic)): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. On the folded fabric, lay the fold edge exactly on the fabric fold, pin it and cut through both layers; opened out you get the whole piece. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

Validation report

Sewability check

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	66.0 cm	66.0 cm	+0.0 mm	□ OK	
Seam pairs	Inseam front ↔ back	41.7 cm	41.7 cm	+0.0 mm	□ OK	
Seam pairs	Front crotch seam (left ↔ right piece)	26.9 cm	26.9 cm	+0.0 mm	□ OK	
Seam pairs	Back crotch seam (left ↔ right piece)	31.3 cm	31.3 cm	+0.0 mm	□ OK	
Seam pairs	Waistband ↔ pants top edge	83.0 cm	83.0 cm	+0.0 mm	□ OK	
Seam pairs	Leg cuff ↔ leg opening (stretched)	70–85 % of 28.5 cm	20.0 cm = 70 %	-85.5 mm	□ OK	rib band is stretched while sewing
Angles/transitions	Centre front ↔ waistband	90°	90.0°	+0.0°	□ OK	
Angles/transitions	Centre back ↔ waistband	90°	90.0°	-0.0°	□ OK	
Angles/transitions	Crotch ↔ inseam (front + back)	180°	180.9°	+0.9°	□ OK	
Angles/transitions	Inseam ↔ hem front	90°	90.1°	+0.1°	□ OK	
Angles/transitions	Hem ↔ side seam front	90°	90.1°	+0.1°	□ OK	
Angles/transitions	Inseam ↔ hem back	90°	90.2°	+0.2°	□ OK	
Angles/transitions	Hem ↔ side seam back	90°	90.2°	+0.2°	□ OK	

Area	Check	Target	Actual	Δ	Result	Note
Angles/transitions	Side seam ↔ waistband back	90°	90.1°	+0.1°	□ OK	
Angles/transitions	Side seam ↔ waistband (front + back)	180°	180.1°	+0.1°	□ OK	
Measurements	Hip circumference pattern = body + ease	85.8 cm	85.8 cm	+0.0 %	□ OK	body 66.0 cm + +30 % ease
Measurements	Top edge fits over the hip	≥ 67.3 cm	95.4 cm	+281.1 mm	□ OK	top edge incl. usable stretch ≥ hip
Measurements	Front rise plausible	14.6 cm – 31.4 cm	27.3 cm	+0.0 mm	□ OK	body rise 18.0 cm, harem drop 7.3 cm
Measurements	Inseam length plausible	39.0 cm – 60.5 cm	55.5 cm	+0.0 mm	□ OK	body inseam 54.0 cm
Measurements	Foot fits through the leg opening	≥ 24.1 cm	29.5 cm	+54.8 mm	□ OK	heel-instep circumference ≈ 1.35 × ankle
Geometry	Front leg: outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 1.0 cm included
Labels	Front leg: grainline & cutting	present	cut 2 (mirrored pair)		□ OK	Main fabric
Geometry	Back leg: outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 1.0 cm included
Labels	Back leg: grainline & cutting	present	cut 2 (mirrored pair)		□ OK	Main fabric
Geometry	Waistband (elastic): outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 1.0 cm included
Labels	Waistband (elastic): grainline & cutting	present	cut 1 on fold		□ OK	Main fabric
Geometry	Leg cuff: outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 1.0 cm included
Labels	Leg cuff: grainline & cutting	present	cut 2		□ OK	Rib knit
Geometry	Waistband (elastic): cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.0° off 90°		□ OK	Centre front
Labels	Front leg: notches	on the cutting line	3 notch mark(s)		□ OK	
Labels	Back leg: notches	on the cutting line	6 notch mark(s)		□ OK	

Result: 31 OK, 0 warning(s), 0 error(s): the pattern is sewable.

Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 858}}`).

Name	Value	Derived from	Note
`hip_width`	85.8 cm	hip × (1 + 30 % ease)	
`front_back_offset`	0.5 cm	front quarter narrower, back quarter wider by this amount	
`front_width`	20.9 cm	hip_width / 4 − front_back_offset	
`back_width`	21.9 cm	hip_width / 4 + front_back_offset	
`front_crotch_extension`	4.3 cm	5 % of hip_width	

Name	Value	Derived from	Note
`back_crotch_extension`	8.6 cm	10 % of hip_width	
`rise`	24.3 cm	(side length − inseam) + seat ease + harem drop − waistband height	
`inseam_finished`	48.2 cm	inseam − allowance to the floor − harem drop + blousing	
`hem_width`	28.5 cm	cuff length / cuff stretch ratio	

Cutting layout · Harem pants size 122

